BANQUET HALL BOOKING SYSTEM

Smart Venue Reservation & Event Management Platform

Phase 6 — Functional & Performance Testing	Jest | Cypress | JMeter | OWASP ZAP

6.1 Testing Overview

Comprehensive testing was conducted across functional, performance, security, and usability dimensions to ensure system reliability and quality. The testing strategy followed a layered approach from unit tests to end-to-end and load testing.

Unit Tests	Integration Tests	Performance Metrics
142 test cases written. 138 passed (97.2%). 4 minor edge cases fixed post-testing.	34 API endpoint tests. All CRUD operations, authentication flows, and payment webhooks tested.	Avg. page load: 1.4s. Search API response: 0.8s. Booking confirmation: 1.2s end-to-end.

Testing Tools

Tool	Purpose
Jest	Unit testing for React components and Node.js business logic (142 test cases).
Cypress	End-to-end (E2E) testing of complete user flows in a real browser environment.
JMeter	Load and performance testing — simulated 500 concurrent users.
OWASP ZAP	Security testing — SQL injection, XSS, and authentication vulnerability scans.
Postman	API testing for all 34 REST endpoints with automated request collections.

Test Cases Summary

TC ID	Test Scenario & Expected Result	Status
TC-01	Verify user registration with valid and invalid credentials — Expected: validation errors shown for invalid inputs, successful account creation for valid data.	✅ Pass
TC-02	Hall search with filters (date: 2025-12-25, capacity: 200, city: Chennai) — Expected: only matching halls returned within 800ms.	✅ Pass
TC-03	Attempt to book an already-reserved hall — Expected: system blocks booking and shows conflict message with next available dates.	✅ Pass
TC-04	Payment flow with valid card — Expected: payment confirmed, booking ID generated, confirmation email sent within 30 seconds.	✅ Pass
TC-05	Load test: 500 concurrent users searching and browsing halls — Expected: response time < 2s, zero server errors.	✅ Pass
TC-06	Security test: SQL injection attempt on search input — Expected: input sanitized, no database exposure, error logged.	✅ Pass
TC-07	Mobile responsiveness on iOS Safari and Android Chrome — Expected: all elements render correctly, no overflow or clipping.	✅ Pass
TC-08	Session timeout after 30 minutes of inactivity — Expected: user redirected to login page with session expiry message.	✅ Pass

Performance Benchmarks

Metric	Target	Actual Result
Avg. Page Load Time	< 2.0 seconds	**1.4 seconds** ✅
Search API Response	< 1.0 second	**0.8 seconds** ✅
Booking Confirmation	< 2.0 seconds	**1.2 seconds** ✅
Concurrent Users (Load)	500 users, 0 errors	**500 users, 0 errors** ✅
Unit Test Pass Rate	≥ 95%	**97.2% (138/142)** ✅
API Test Coverage	100% of endpoints	**34/34 endpoints** ✅